

25D Series — Metal Oxide Varistors

25 mm round, epoxy-encapsulated MOVs with radial leads · Datasheet · Rev. 2026-10-21

Description

The 25D series is a line of 25 mm round, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs. Rated In 10 kA / Imax 20 kA (8/20 µs), the range covers varistor voltages from 100 V to 820 V in 22 model grades. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.



Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- 25 mm round MOVs, epoxy encapsulated, with radial leads
- 22 model grades — varistor voltage from 100 V to 820 V
- In 10 kA, Imax 20 kA (8/20 µs)
- Low-residual-voltage range available for tighter clamping
- Leaded disc format for PCB and point-to-point assembly
- Discs pressed, sintered, metallized and tested in our own plant

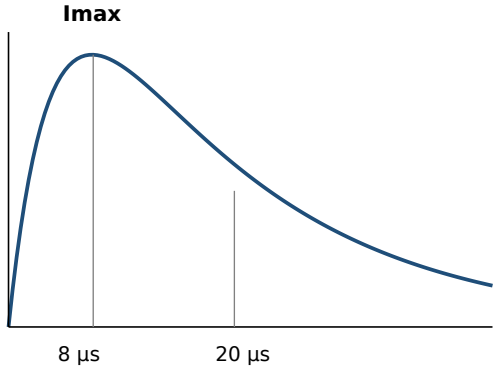
Agency approvals

Column definitions: U1mA = varistor voltage at DC 1 mA; ΔU1mA = varistor-voltage tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In, shown as reference value–ratio; ETM = maximum energy absorption (2 ms). The catalog prints UL / C-UL / TÜV columns for this table but leaves them blank for every model in this series, so they are omitted here. Notes: 25DK181s ETM prints as 140 J in the catalog — below the 160 J printed for 25DK151 — shown as printed.
For series not yet UL recognized, we can pursue UL certification for your program. Full certification coverage for all series is on the Certifications page at www.flyboatmov.com.

Absolute maximum ratings

Parameter	Range / limit	Unit
Continuous operating voltage, AC (VM(AC)rms)	60 to 510	V
Continuous operating voltage, DC (VM(DC))	85 to 670	V
Nominal discharge current In (8/20 µs)	10	kA
Maximum discharge current Imax (8/20 µs)	up to 20	kA
Maximum energy absorption (2 ms)	up to 720	J

Impulse current waveform 8/20 µs (definition)



CAUTION: stresses above the ratings in the tables below may cause permanent damage. The ranges above summarize the per-model ratings in the sections below; always design to the row for the exact model code.

Model naming and standard marking

Model naming: 25D K 431 — 25 = nominal size (mm), D = round disc; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default). Standard marking: the manufacturer logo and the model code are etched on the encapsulation.

1. Electrical characteristics — standard ratings

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
25DK101	100	±10	60	85	10	20	360-3.6	90
25DK121	120	±10	75	100	10	20	400-3.6	110
25DK151	150	±10	95	125	10	20	450-3.0	160
25DK181	180	±10	115	150	10	20	490-2.8	140
25DK201	200	±10	130	170	10	20	520-2.6	200
25DK221	220	±10	140	180	10	20	550-2.6	220
25DK241	240	±10	150	200	10	20	600-2.6	235
25DK271	270	±10	175	225	10	20	680-2.6	270
25DK301	300	±10	195	250	10	20	770-2.6	300
25DK331	330	±10	210	270	10	20	830-2.55	330
25DK361	360	±10	230	300	10	20	910-2.55	370
25DK391	390	±10	250	320	10	20	1000-2.55	410
25DK431	430	±10	275	350	10	20	1050-2.45	460
25DK471	470	±10	300	385	10	20	1100-2.4	500
25DK511	510	±10	320	410	10	20	1150-2.35	530
25DK561	560	±10	350	450	10	20	1250-2.35	580
25DK621	620	±10	385	505	10	20	1550-2.35	630
25DK681	680	±10	420	560	10	20	1650-2.35	640
25DK711	710	±10	440	585	10	20	1700-2.35	650
25DK751	750	±10	460	615	10	20	1700-2.3	670
25DK781	780	±10	485	640	10	20	1750-2.3	690
25DK821	820	±10	510	670	10	20	1850-2.3	720

2. Electrical characteristics — low-residual-voltage ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
25DK431	430	±10	275	350	10	20	990-2.2	460

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
25DK471	470	±10	300	385	10	20	1070-2.2	500
25DK511	510	±10	320	410	10	20	1150-2.2	530
25DK561	560	±10	350	450	10	20	1250-2.15	580
25DK621	620	±10	385	505	10	20	1350-2.1	630
25DK681	680	±10	420	560	10	20	1520-2.1	640
25DK711	710	±10	440	585	10	20	1600-2.1	650
25DK751	750	±10	460	615	10	20	1700-2.1	670
25DK781	780	±10	485	640	10	20	1760-2.1	690
25DK821	820	±10	510	670	10	20	1800-2.08	720

Ordering and support: samples and design-in support — North America: Thomas Dong, Flyboat Electronic LLC, Bellevue, WA (thomasd@flyboatmov.com); China: Colin Fei (colinf@flyboatmov.com). Specifications are subject to change without notice; verify critical ratings against the model row above before design freeze.